

Week 11 — UI/Formatting Output

Goal

Improve readability and consistency of UI and trail formatting.

Required Outputs

- ☐ UI polish: aligned labels/inputs, consistent spacing
- ☐ Trail formatting standardized (clear headings and spacing)
- ☐ Before/After evidence provided

Evidence to Attach

- ☐ Before/After screenshots (UI + trail)

Reflection (Write short answers)

- Which UI change improved understanding the most?

EVIDENCE TO ATTACH

UI BEFORE

The screenshot shows the Newton-Raphson Root Finder v2.0 (CustomTkinter) application in a light theme. The window has a blue title bar and a white background. The main content is divided into two panels: "Solver Configuration" on the left and "Solution Trail" on the right. The "Solver Configuration" panel contains a "Method" dropdown menu with "Newton-Raphson" selected, a "Function" dropdown menu with "x^2-2" selected, an "Initial Guess" input field with "1.5", a "Tolerance" input field with "1e-6", and a "Max Iterations" input field with "50". Below these fields are three buttons: "Compute" (blue), "Clear" (gray), and "Export" (green). At the bottom of the "Solver Configuration" panel is a "Final Root:" label. The "Solution Trail" panel is empty. The bottom of the window has a "Theme:" dropdown menu set to "System".

Newton-Raphson Root Finder v2.0 (CustomTkinter)

Newton-Raphson Root Finder

Solver Configuration

Method:
 Newton-Raphson
 Secant
 x^2-2

Initial Guess:
 1.5

Tolerance:
 1e-6

Max Iterations:
 50

Compute Clear Export

Final Root:

Theme: System

The screenshot shows the Newton-Raphson Root Finder v2.0 (CustomTkinter) application in a dark theme. The window has a dark blue title bar and a dark gray background. The main content is divided into two panels: "Solver Configuration" on the left and "Solution Trail" on the right. The "Solver Configuration" panel contains a "Method" dropdown menu with "Newton-Raphson" selected, a "Function" dropdown menu with "cos(x) - x" selected, an "Initial Guess" input field with "10", a "Tolerance" input field with "1e-10", and a "Max Iterations" input field with "3". Below these fields are three buttons: "Compute" (blue), "Clear" (gray), and "Export" (green). At the bottom of the "Solver Configuration" panel is a "Final Root:" label with the value "-20202.9263429". The "Solution Trail" panel contains a text area with the following text:
 GIVEN:
 Method: Newton-Raphson
 Function: cos(x) - x
 Initial guess: 10
 Tolerance: 1e-10
 Max iterations: 3
 VALIDATION:
 1. All inputs valid.
 METHOD:
 Newton-Raphson
 STOPPING RULES:
 1. Stop if $|x_{new} - x| < \text{tolerance}$ (1e-10)
 2. Stop if iterations exceed max (3)
 3. Stop if derivative = 0
 STEPS:
 2. i=1, x=10.0000000000, f(x)=-1.084e+01, x_next=-13.7709942015, err=2.377e+01
 3. i=2, x=-13.7709942015, f(x)=1.413e+01, x_next=199.3511775153, err=2.131e+02
 4. i=3, x=199.3511775153, f(x)=-1.995e+02, x_next=-20202.9263428707, err=2.040e+04
 FINAL ANSWER:
 x = -20202.9263429
 STOPPING CONDITION:
 STOP: Reached max iterations (3)
 SUMMARY:
 Iterations: 3
 Time: 0.0011s
 Timestamp: 2026-03-19 18:17:15

Newton-Raphson Root Finder v2.0 (CustomTkinter)

Newton-Raphson Root Finder

Solver Configuration

Method:
 Newton-Raphson

Function:
 cos(x) - x

Initial Guess:
 10

Tolerance:
 1e-10

Max Iterations:
 3

Compute Clear Export

Final Root:
 -20202.9263429

Solution Trail

GIVEN:
 Method: Newton-Raphson
 Function: cos(x) - x
 Initial guess: 10
 Tolerance: 1e-10
 Max iterations: 3

VALIDATION:
 1. All inputs valid.

METHOD:
 Newton-Raphson

STOPPING RULES:
 1. Stop if $|x_{new} - x| < \text{tolerance}$ (1e-10)
 2. Stop if iterations exceed max (3)
 3. Stop if derivative = 0

STEPS:
 2. i=1, x=10.0000000000, f(x)=-1.084e+01, x_next=-13.7709942015, err=2.377e+01
 3. i=2, x=-13.7709942015, f(x)=1.413e+01, x_next=199.3511775153, err=2.131e+02
 4. i=3, x=199.3511775153, f(x)=-1.995e+02, x_next=-20202.9263428707, err=2.040e+04

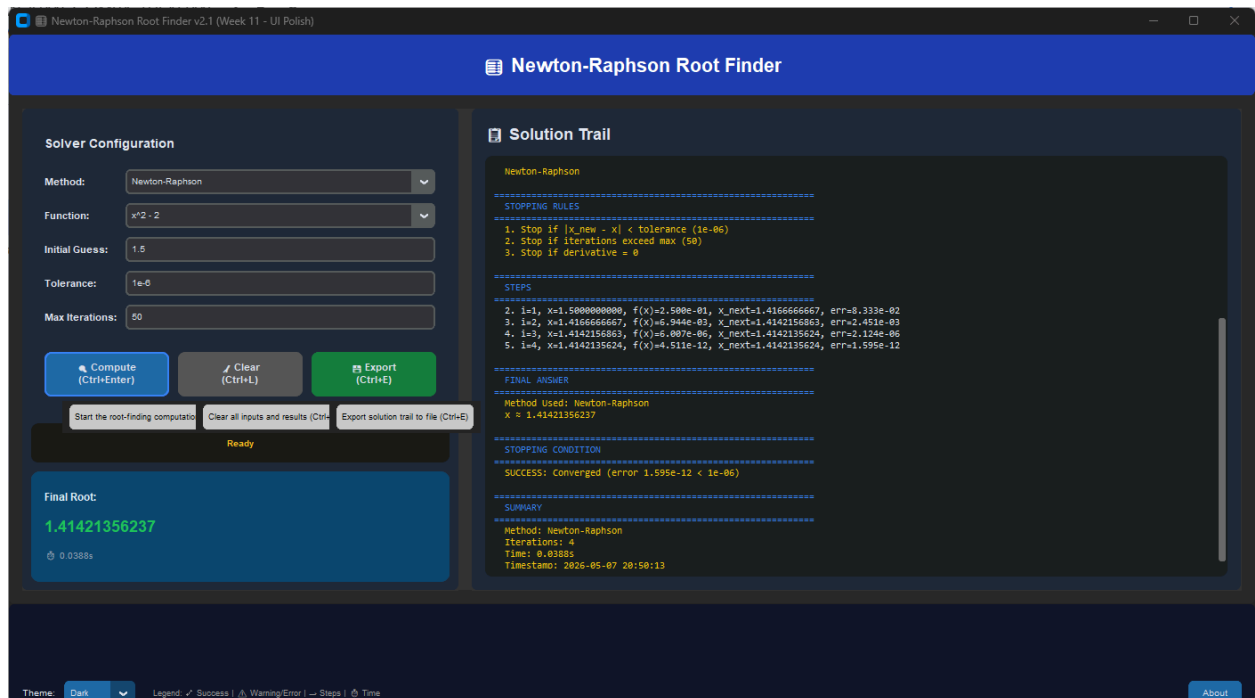
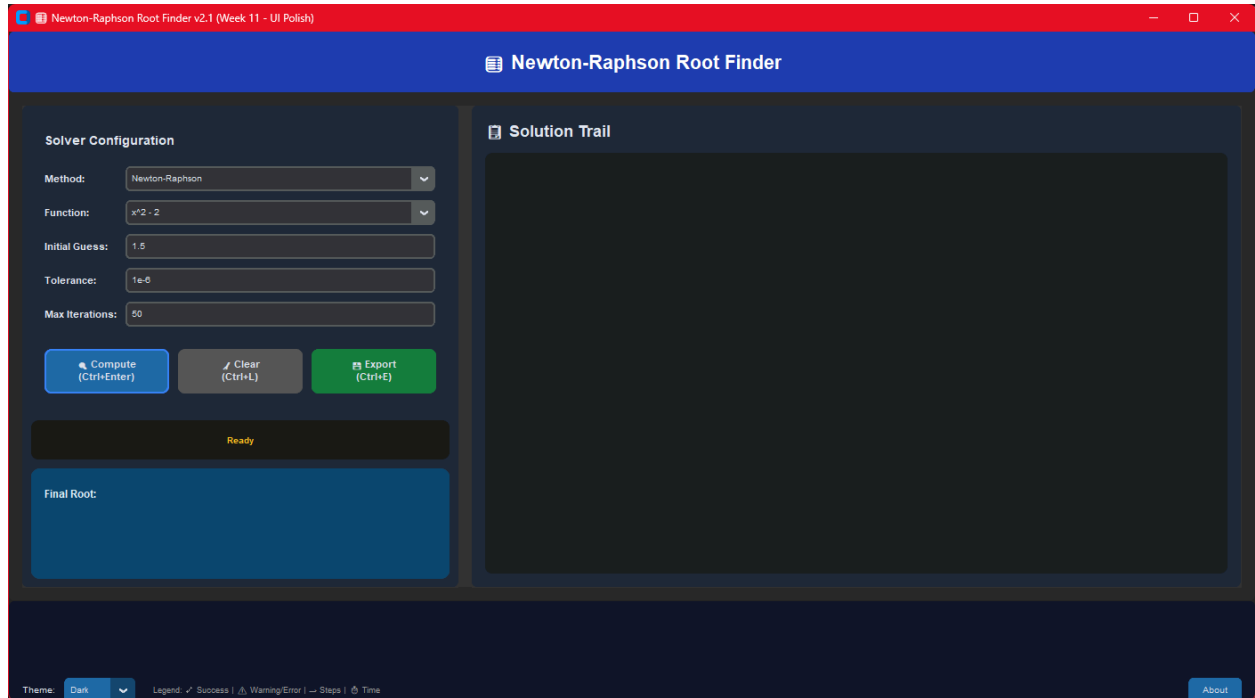
FINAL ANSWER:
 x = -20202.9263429

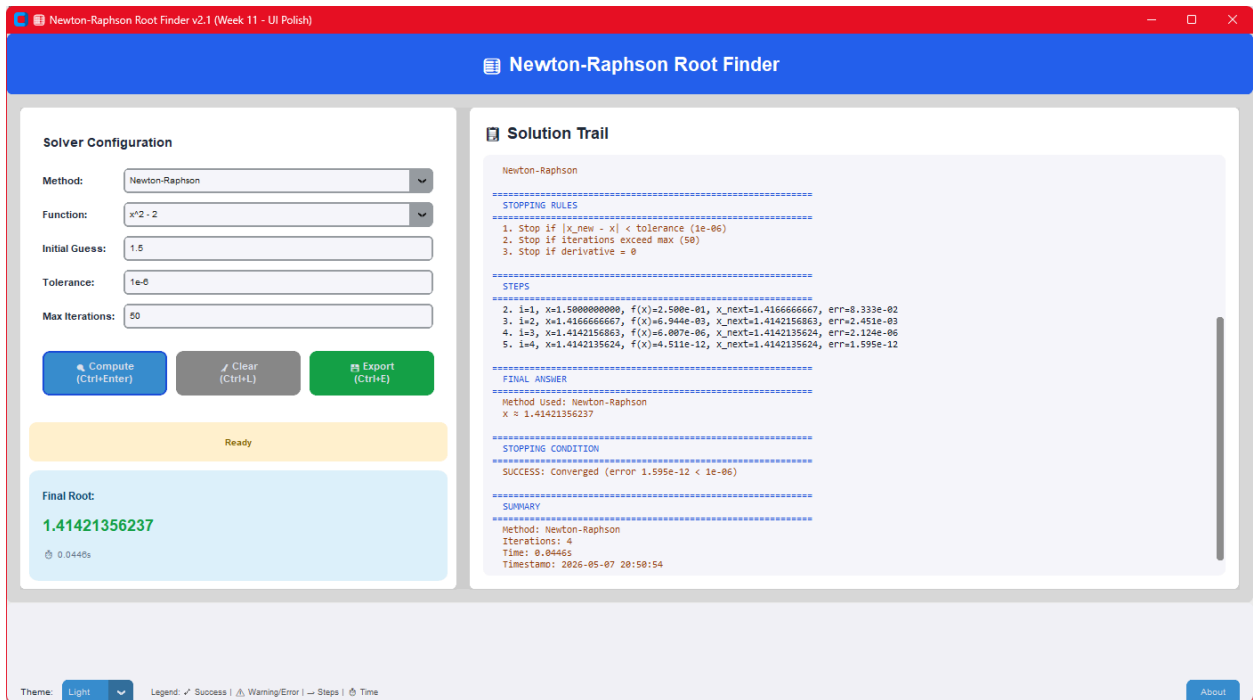
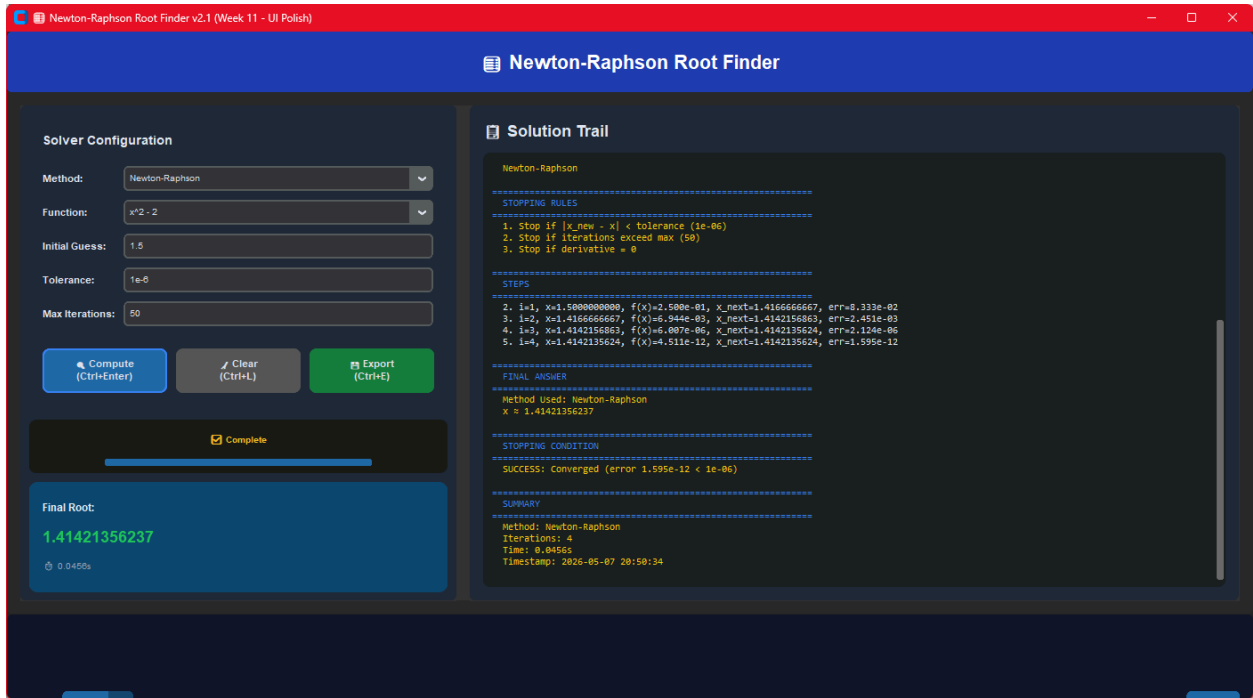
STOPPING CONDITION:
 STOP: Reached max iterations (3)

SUMMARY:
 Iterations: 3
 Time: 0.0011s
 Timestamp: 2026-03-19 18:17:15

Theme: Dark

UI AFTER





Reflection

Which UI change improved understanding the most?

The most impactful UI change was standardizing the solution trail with clear section headings and spacing, because it turned the entire computation flow into a readable narrative GIVEN, VALIDATION, EDGE CASE ANALYSIS, METHOD, STEPS, FINAL ANSWER, and SUMMARY and the added separators and icons like ✓, △, and → make it easy to scan and understand what happened.

Weekly Sign-off

Group Leader Name/Signature:  CHARLES RUSSEL O. SAN JUAN Date: 5/7/2026

Member Name/Signature:  CHRISTINA GOMBA Date: 5/7/2026

Member Name/Signature:  JOHN MICHAEL TULALIAN Date: 5/7/2026